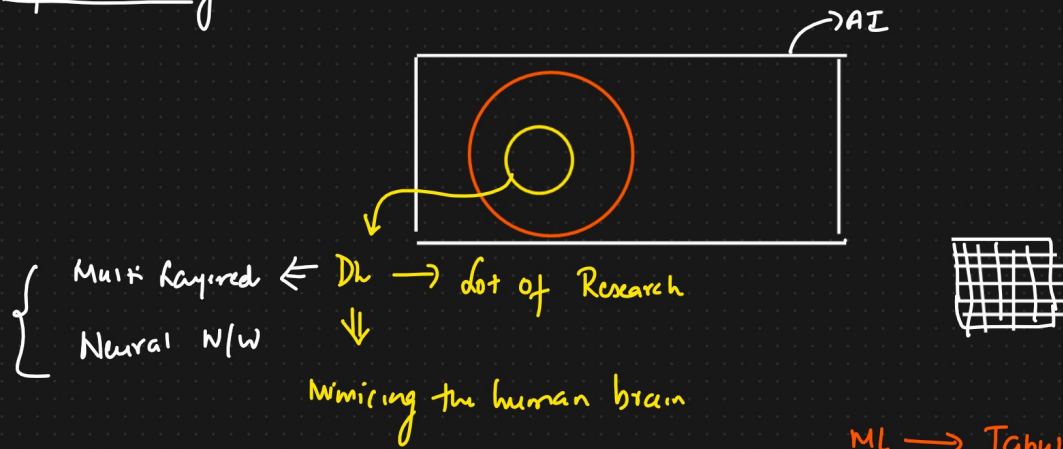


Deep Learning



Deep Learning

- ① ANN $\rightarrow$ Artificial Neural N/W $\rightarrow$ CLASSIFICATION, Regression
- ② CNN $\rightarrow$ Convolutional Neural N/W $\rightarrow$ IP = Images, Video frames $\rightarrow$ VGG16, Resnet
- $\Downarrow$
- Computer Vision
- Eg: RCNN, Masked RCNN, Detection, YOLOv6
- Yolo v8
- ③ RNN $\rightarrow$ Recurrent Neural N/W $\rightarrow$ IP $\Rightarrow$ Text Data, Time Series Data.
- $\rightarrow$ LSTM RNN, RNN GRU, Bidirectional LSTM RNN, Encoder Decoder, Transformers, BERT, Attention Models
- $\Downarrow$
- Tensorflow, PyTorch

Why Deep Learning Is Becoming Popular?

2005 $\rightarrow$ Facebook, Orkut $\rightarrow$ Social media platform



DATA WAS BEING GENERATED

2011 $\Rightarrow$ BIG DATA

EXPONENTIALLY. $\Rightarrow$ HADOOP $\Rightarrow$ Unstructured data, Structured data



STORING THESE DATA EFFICIENTLY



Images, Text, Documents
Videos

S3 Buckets

2011-2011 $\Rightarrow$ Cloudera, Hoshinworks



2015



Can we use the data to make the product better



DATA SCIENCE



Deep Learning

NVIDIA



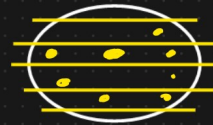
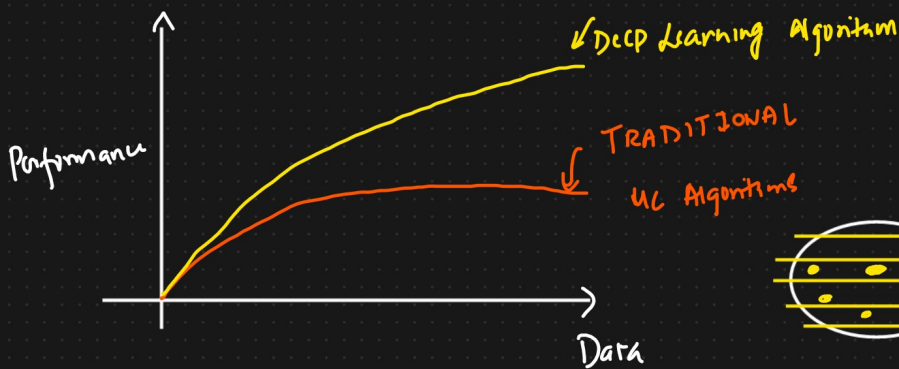
TITAN RTX

RTX 3080

① Hardware Requirement $\Rightarrow$ GPUs $\Rightarrow$ Nvidia GPUs

Price $\downarrow$

② Huge amount of data is generated $\Rightarrow$ Deep learning model performs well



Tumor

Brain Scan

③ Deep Learning is been used in Many Domains

3D Image

① Medical $\rightarrow$ Prediction of diseases, X-rays, Bone crack, lungs disease.

MRI Scan

DATA

② Ecommerce

③ Retail

④ Logistic

① Perception [Artificial Neuron or Neural Network Unit]



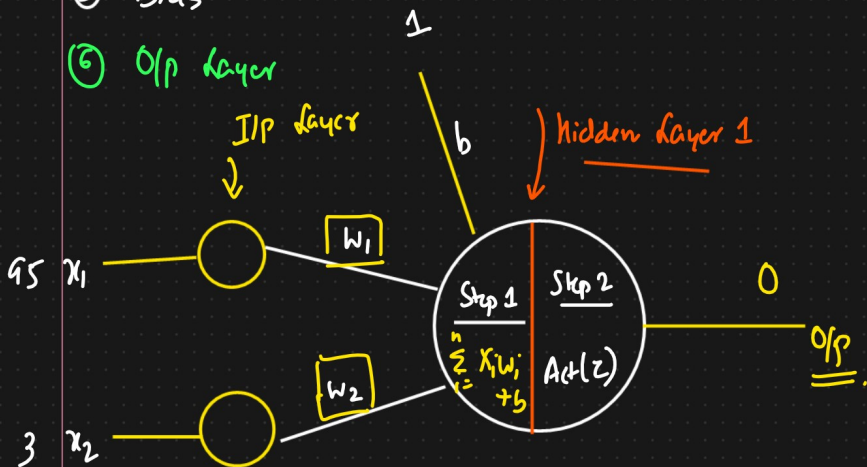
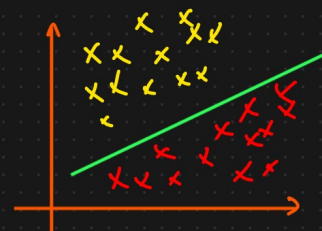
- ① I/P layer
- ② Hidden layer
- ③ Weights
- ④ Activation function
- ⑤ Bias
- ⑥ O/P layer

Single Layered Neural N/W

[Binary Classification]



Linear Separable



DATASET

x_1	x_2	y
<u>IQ</u>	<u>No. of study hours</u>	<u>O/P PASS/FAIL</u>
95	3	0
110	4	1
100	5	1

Step 1

Feed Forward Neural N/W → Forward Propagation

$$Z = x_1 w_1 + x_2 w_2 + 1 \times b \quad \text{Intercept}$$

$$Z = \sum_{i=1}^n x_i w_i + b$$

$$Z = -2$$

$$Z = 5 \Rightarrow 1$$

$$5 > 0$$

Step Function

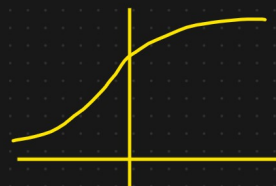


Threshold = 0

$$0.5$$

0 or 1

Sigmoid



Hyperparameter

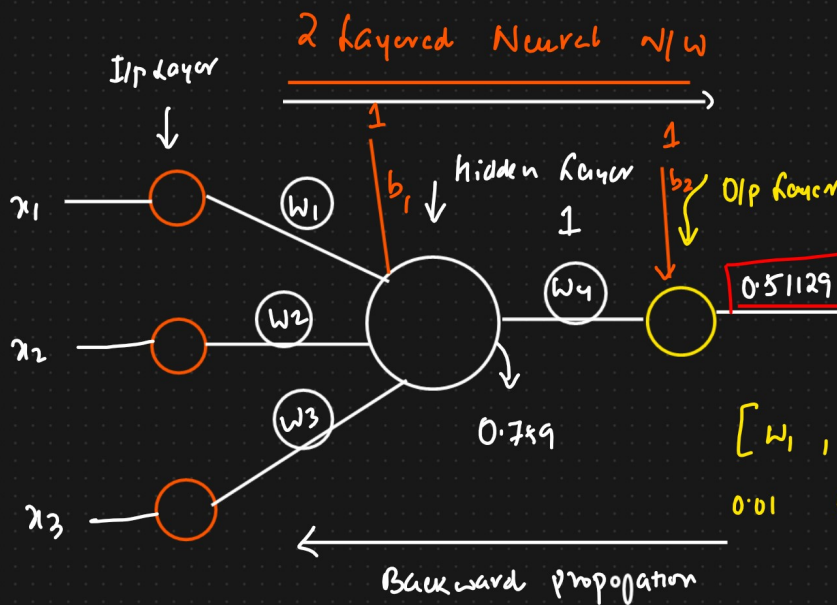
$$0.1, 0.2, 0.3, 0.4, 0.5$$

Multi Layered Perceptron Model [ANN]

- ① Forward Propagation
- ② Backward Propagation
- ③ Loss function
- ④ Activation function
- ⑤ Optimizers.

Binary

x_1	x_2	x_3	O/p
IQ	Study hours	Play hours	
→ 95	4	4	1
100	5	2	1
95	2	7	0



$$\text{Loss} = 1 - 0.51129 = 0.48871$$

$$(y - \hat{y})$$

↓ Initialized

$$[w_1, w_2, w_3] \quad [0.01]$$

$$0.01 \quad 0.02 \quad 0.03$$

HL 1

$$\text{Step 1} = 95 \times 0.01 + 4 \times 0.02 + 4 \times 0.03 + 0.01$$

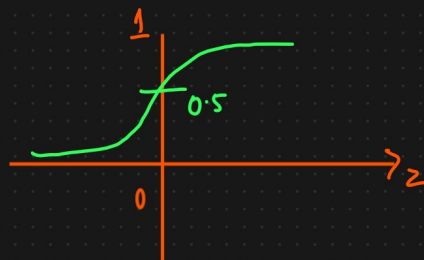
$$Z = 1.151$$

$$\text{Step 2} = \text{Activation}(Z)$$

$$= \frac{1}{1 + e^{-(1.151)}} = 0.759$$

Sigmoid

$$y = \frac{1}{1 + e^{-z}}$$



HL 2 :

$$Z = 0.759 \times 0.02 + 1 \times 0.03$$

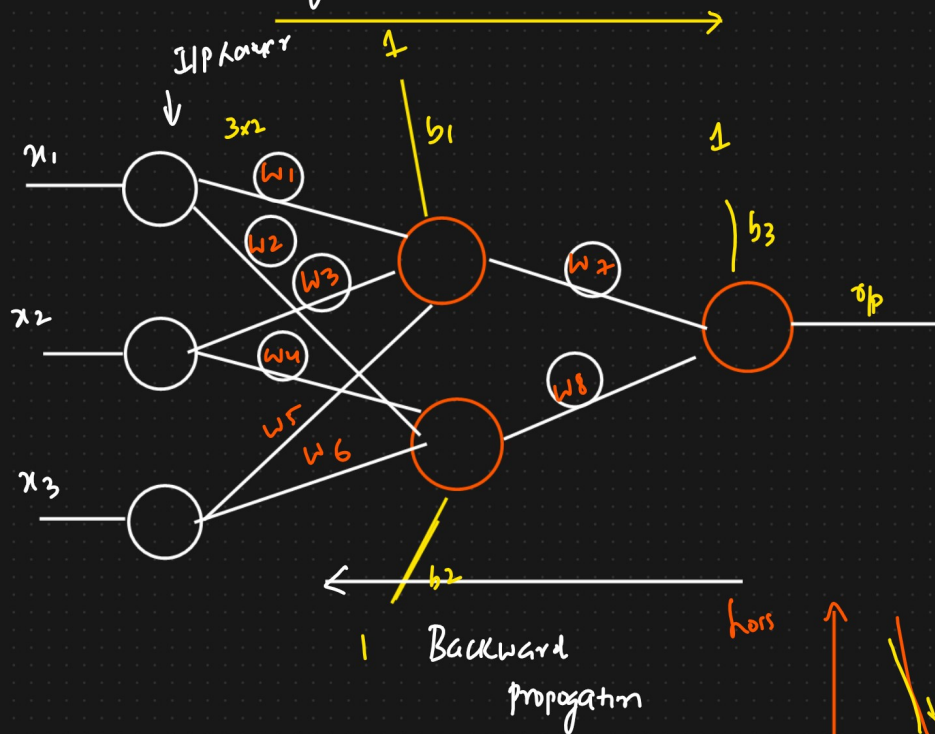
$$= 0.04518$$

$$w_4 = [0.02] \quad b_2 = 0.03$$

Step 2 :

$$O_2 = \frac{1}{1 + e^{-0.04519}} = 0.51129 \Rightarrow \underline{\underline{O/p \text{ Layer}}}$$

① Backward Propagation And Weight Update Formula



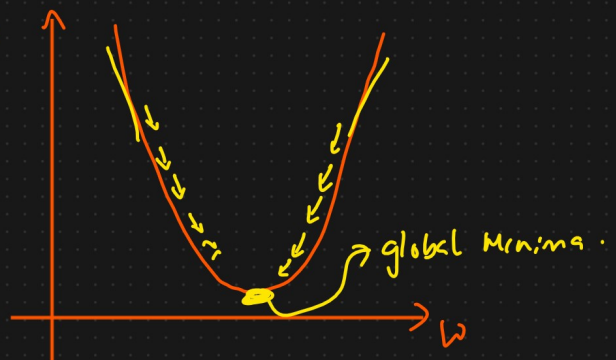
Loss function
 $(y - \hat{y})^2 \downarrow \downarrow \downarrow \downarrow$

Optimizers $\Rightarrow$ Gradient Descent

Weight update Formula

$$w_{\text{new}} = w_{\text{old}} - \eta \left[\frac{\partial L}{\partial w_{\text{old}}} \right]$$

$\hookrightarrow$ slope



$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w_{\text{old}}}$$

$\Rightarrow$ Weight Update Formula.

$$b_{\text{new}} = b_{\text{old}} - \eta \frac{\partial L}{\partial b_{\text{old}}}$$

$\Rightarrow$ bias Update Formula.

Optimizers : To reduce the loss value

$$W_{new} = W_{old} - \eta (-ve)$$

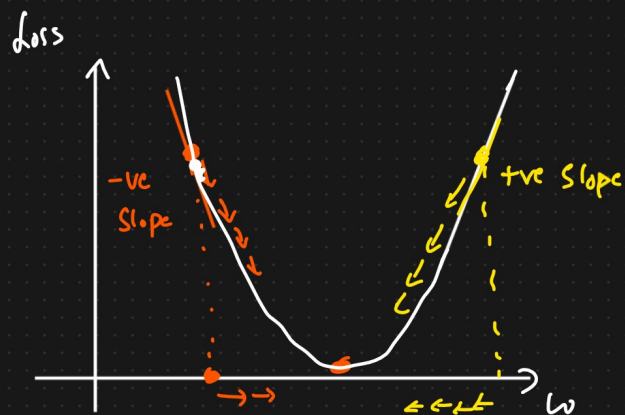
η = learning Rate

$$W_{new} = W_{old} + (+ve)$$

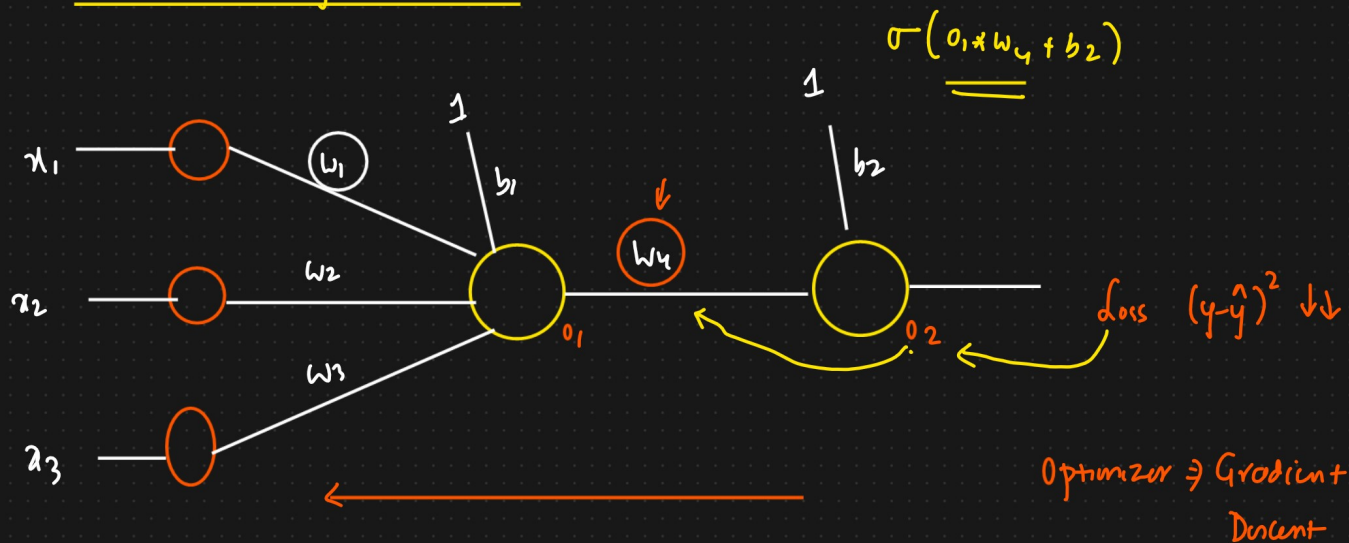
$$W_{new} >> W_{old}$$

$$W_{new} = W_{old} - \eta (+ve)$$

$$W_{new} << W_{old}$$



⑧ Chain Rule of Derivative



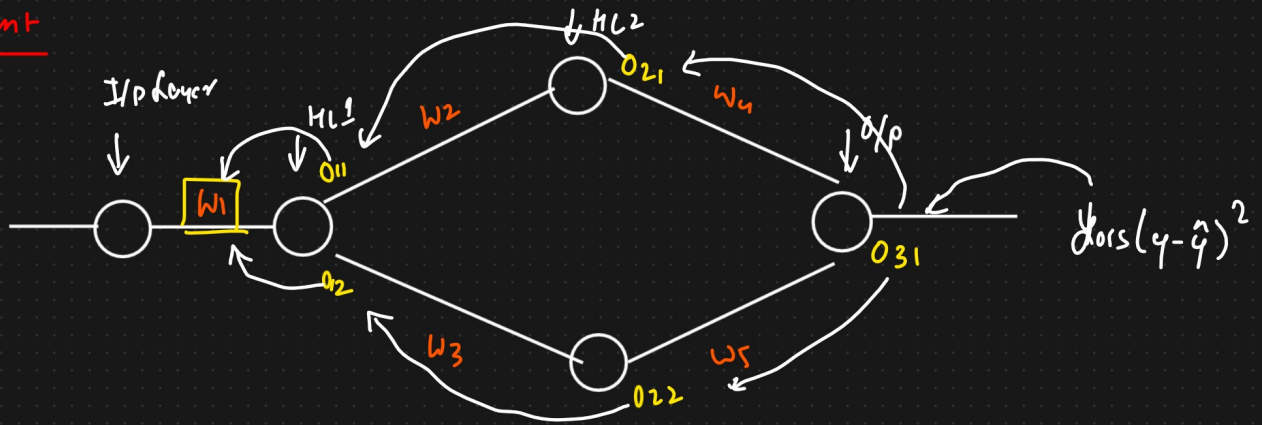
$$W_{4_{new}} = W_{4_{old}} - \eta \left[\frac{\partial h}{\partial W_{4_{old}}} \right] \Rightarrow \text{slope}$$

$$\frac{\partial h}{\partial W_{4_{old}}} = \frac{\partial h}{\partial o_2} * \frac{\partial o_2}{\partial W_{4_{old}}} \quad \left\{ \text{Chain Rule of Derivation} \right\}$$

$$W_{new} = W_{old} - \eta \left[\frac{\partial h}{\partial W_{old}} \right]$$

$$\frac{\partial h}{\partial W_{1_{old}}} = \frac{\partial h}{\partial o_2} * \frac{\partial o_2}{\partial o_1} * \frac{\partial o_1}{\partial W_{1_{old}}}$$

Assignment



$$w_{1\text{new}} = w_{1\text{old}} - \eta \left[\frac{\partial L}{\partial w_{1\text{old}}} \right]$$

$$\frac{\partial L}{\partial w_{1\text{old}}} = \left[\frac{\partial L}{\partial o_{31}} * \frac{\partial o_{31}}{\partial o_{21}} * \frac{\partial o_{21}}{\partial o_{11}} * \frac{\partial o_{11}}{\partial w_{1\text{old}}} \right]$$

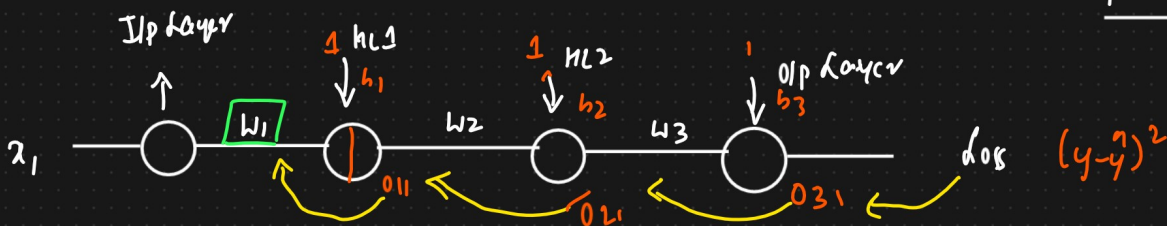
+

$$\left[\frac{\partial L}{\partial o_{31}} * \frac{\partial o_{31}}{\partial o_{22}} * \frac{\partial o_{22}}{\partial o_{12}} * \frac{\partial o_{12}}{\partial w_{1\text{old}}} \right]$$

Dep layered NN $\Rightarrow$ Sigmoid Activation will not work

⑧ Vanishing Gradient Problem And Activation functions

$$\boxed{0 - 0.25} \checkmark$$



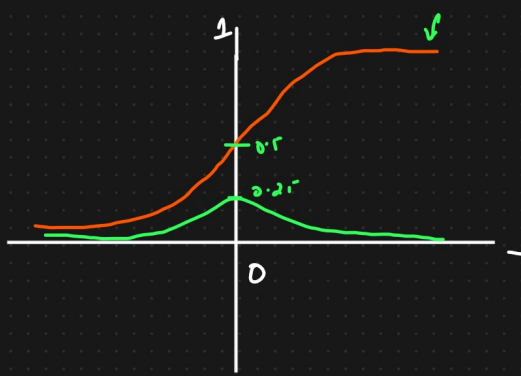
Sigmoid Activation

$$\sigma(z) = \left[\frac{1}{1 + e^{-z}} \right] \rightarrow 0 - 0.25$$

$$\boxed{0 \leq \sigma(z) \leq 1}$$

Derivation of $\sigma(z)$

$$\boxed{0 \leq \sigma(z) \leq 0.25}$$



$$w_{new} = w_{old} - \eta \left[\frac{\partial h}{\partial w_{old}} \right] \quad \begin{matrix} \uparrow 0.01 \\ \downarrow \text{very small} \end{matrix} \quad \left[\frac{w_{new} - w_{old}}{w_{old}} \right] \quad \begin{matrix} \text{Small value} \\ \left[0 - 0.25 \right] \quad \left[0 \leftarrow \right] \end{matrix}$$

$$\frac{\partial h}{\partial w_{old}} = \frac{\partial h}{\partial o_{31}} * \left[\frac{\partial o_{31}}{\partial o_{21}} \right] * \frac{\partial o_{21}}{\partial o_{11}} * \frac{\partial o_{11}}{\partial w_{11}} \quad \begin{matrix} \downarrow \text{small value} \end{matrix}$$

$$o_{31} = \sigma(w_3 * o_{21} + b_3) \quad \boxed{z = w_3 * o_{12} + b_3} \checkmark$$

$$o_{31} = \sigma(z)$$

$$\frac{\partial o_{31}}{\partial o_{21}} = \frac{\partial \sigma(z)}{\partial o_{21}} = \frac{\partial \sigma(z)}{\partial z} * \frac{\partial z}{\partial o_{21}} \quad \{\text{Chain Rule}\}$$

$$0 - 0.25$$



$$\frac{\partial (w_3 * o_{21} + b_3)}{\partial o_{21}} \Rightarrow w_3$$

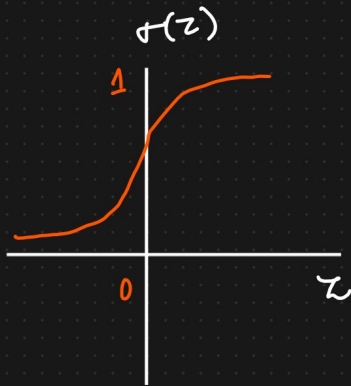
$$\frac{\partial o_{31}}{\partial o_{21}} = [0 - 0.25] * w_3 \Rightarrow \text{small value}$$

① To fix this problem we use other Activation function

- ① Tanh ② ReLU ③ PReLU ④ ELU ⑤ Swiss.

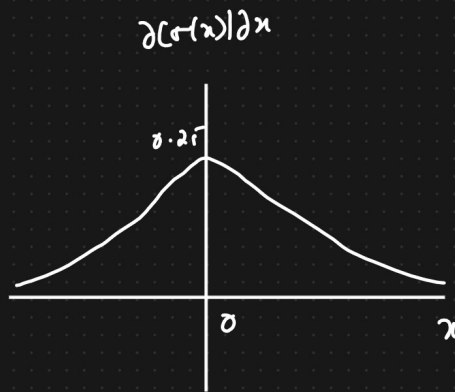
Activation function

① Sigmoid Activation function



Advantages

- ① Clear prediction 1 or 0

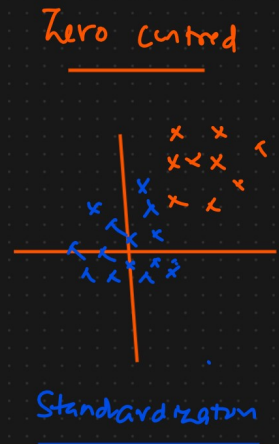


Disadvantages

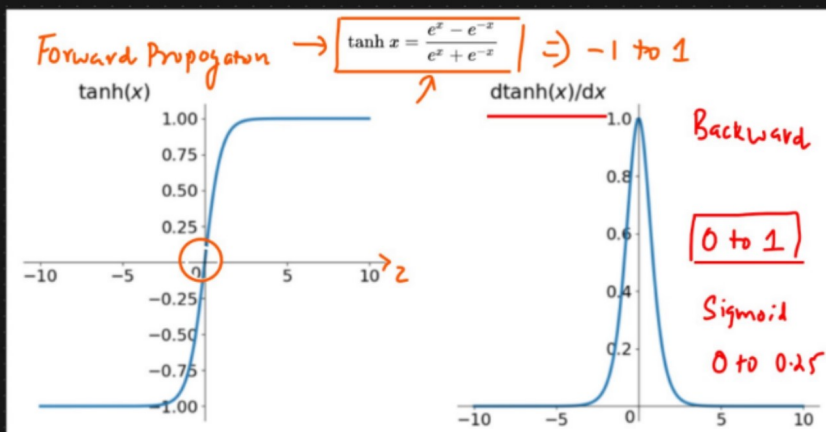
- ① Prone to vanishing gradient problem
- ② Function output is not zero centered

↓

Efficient weight update



② Tanh Activation function



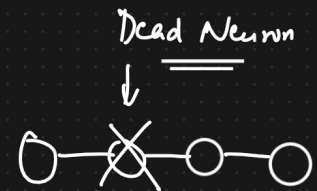
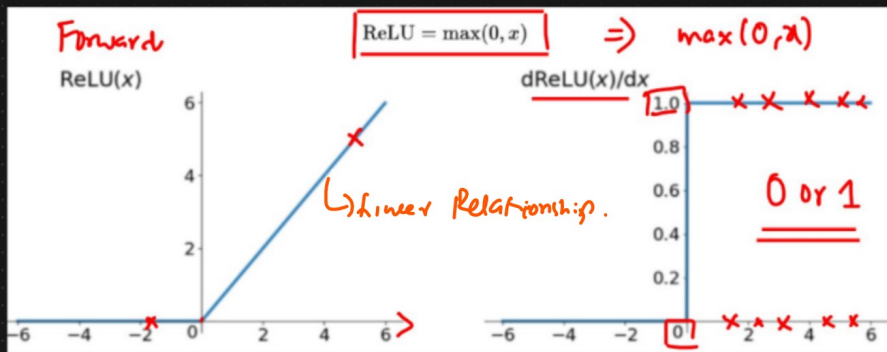
Advantage

- ① Zero centered

Disadvantage

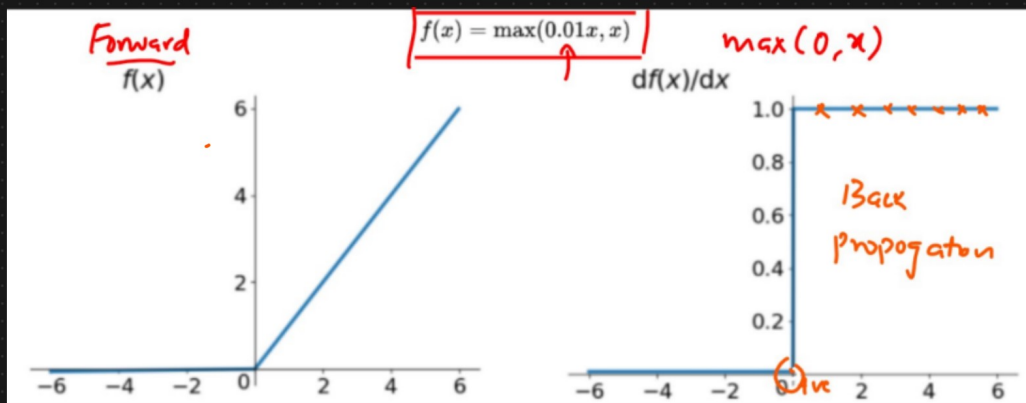
- ① Time consuming
- ② Vanishing gradient problem still exists.

③ Relu Activation Function



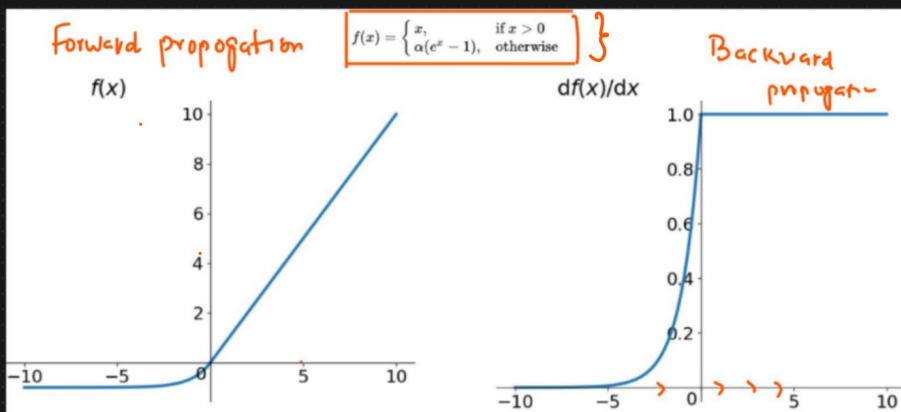
④ Leaky Relu or Parametric Relu

$\max(\alpha x, x)$ $\rightarrow$ Hyperparameter



Prevents
Dead Neuron

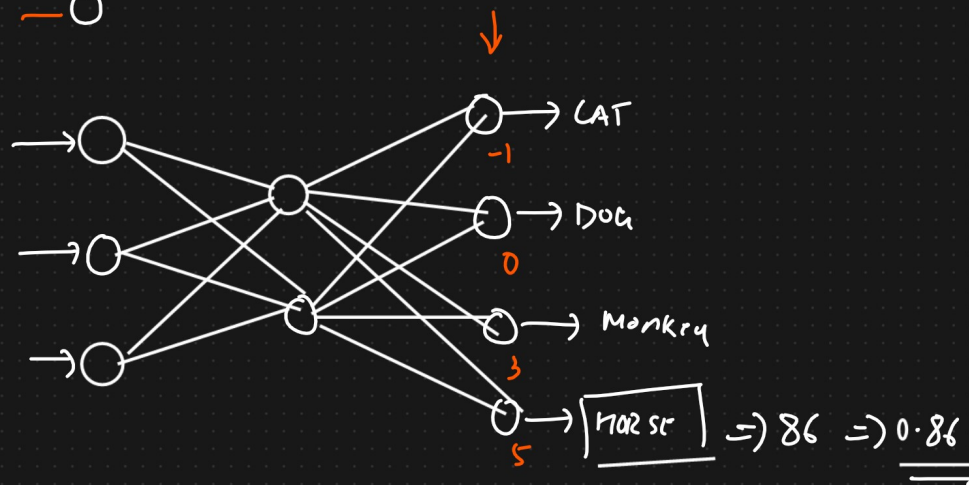
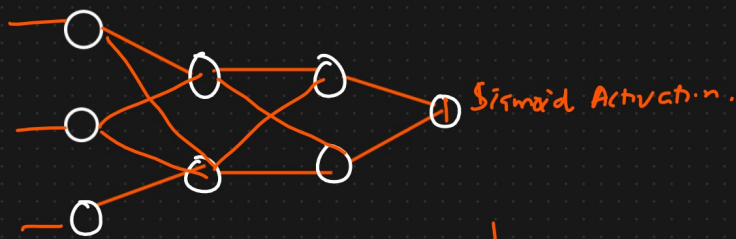
⑤ ELU (Exponential Linear Units)



Disadvantage

Time consuming

⑥ Softmax Activation Function



Softmax Activation

$$\text{Softmax} = \frac{e^{y_i}}{\sum_{k=0}^n e^{y_k}}$$

Softmax Activation $\Leftarrow$ Cat = $\frac{e^{-1}}{e^{-1+0+3+5}} = 0.00033$

Dog = $\frac{e^0}{e^{-1+0+3+5}} = 0.0024$

Monkey = $\frac{e^3}{e^{-1+0+3+5}} = 0.0183$

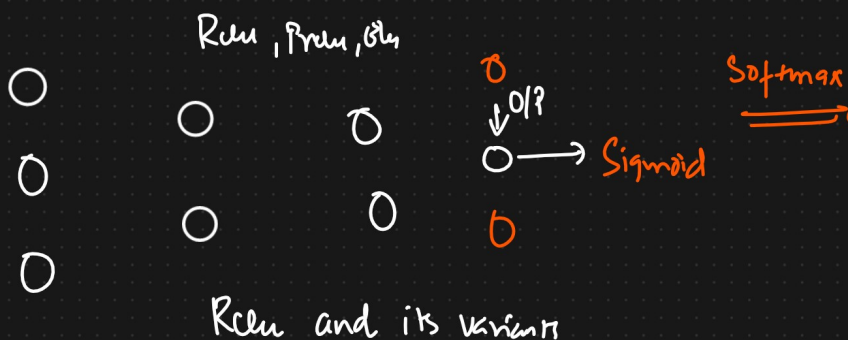
Horse = $\frac{e^5}{e^{-1+0+3+5}} = 0.1353$

Pr(Horse) = $\frac{0.1353}{0.00033 + 0.0024 + 0.0183 + 0.1353}$

$\approx 86\%$

86%

⑦ Which Activation function To use When



Regression => Linear

Sigmoid -> Binary classif

Softmax => Multiclass classification